

Linux shim (kernel WG / tun)

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Volume 4 (Clients) nano-detail specification — deepens the **Linux** row of the TunnelPlatform shim matrix [03-client §5.4]. SPEC ONLY: it pins **what to build** — concrete Rust/Dart signatures, the TunnelPlatform contract impl, FFI surface, the helper-daemon IPC, state machines, lifecycle, privilege model, memory budgets, error handling, edge cases, and test points tied to the §11.4.169 closed test-type vocabulary. It does **not** write the shipping product. Sources cited inline by id — [03-client §N] = final/03-client-core-and-ui.md (the pass-1 client overview this deepens); [01-TXP §N] = final/v02-data-plane/transport-trait.md; [01-ORC §N] = final/v02-data-plane/orchestrator-and-state.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_UI §N] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [04_P0 §N] = 04_VPN_CLD/HelixVPN-Phase0-Spike.md; [SYN §N] = v09-research/_SYNTHESIS.md. [research-flutter_ffi] / [research-ios_android] = the FFI / native-shim pattern digests (referenced for the generic flutter_rust_bridge/UniFFI patterns doc 03 already grounds). Anything not grounded in that evidence base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

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0. Position, ownership, and what makes Linux different

The Linux shim is one row of the TunnelPlatform matrix [03-client §5, 04_ARCH §5.3]: *configure the OS tunnel, hand packets to/from helix-core, report lifecycle — everything else is shared* Dart [04_UI §6]. This document owns **only** the Linux instantiation of that contract; it does not re-specify the FFI surface (owned by [03-client §3]), the Transport trait ([01-TXP]), or the orchestrator/TunnelStatus ([01-ORC]) — it **consumes** them.

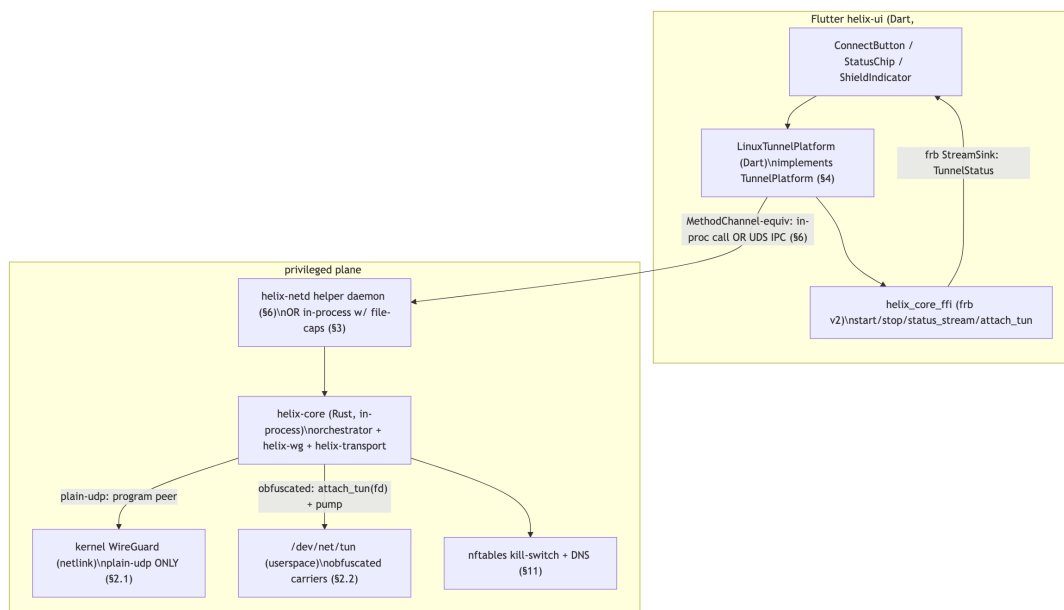
Linux is the **reference / simplest** shim platform [04_P0 §9, 03-client §5.4]: on desktop the core can drive the TUN **in-process** — there is no OS-managed extension process (unlike iOS Network Extension or Android VpnService), no mandatory IPC hop (unlike the Windows privileged service). That simplicity is exactly why Phase-0 gate G5 (flutter_rust_bridge drives the core from Dart) ships on Flutter-Linux first [04_P0 §9].

But Linux carries **two structural complications the other shims do not**:

#	Linux-specific reality	Consequence
L0a	Two real data-path backends exist — kernel WireGuard (netlink) and userspace /dev/net/tun — with <i>different obfuscation reach</i> (§2).	A backend-selection decision the core makes per active Transport, not a fixed choice.
L0b	CAP_NET_ADMIN is mandatory to create a TUN, program routes, and install the kill-switch, yet the Flutter GUI runs unprivileged.	A privilege seam (§7) — either file-caps on a helper or a polkit-gated helper daemon (§6), mirroring the Windows service model.
L0c	Linux is the connector's primary host — the headless <code>--mode=connector</code> daemon (§9) runs	

#	Linux-specific reality	Consequence
	on Linux far more than the GUI does [04_ARCH §5.4, 01-ORC §9].	This doc owns the <i>daemon</i> deployment too, not only the desktop GUI shim.

Ownership boundary (CI1 [03-client §0.1]). The shim owns the OS tunnel *lifecycle and privilege* (create/configure/destroy the TUN, acquire CAP_NET_ADMIN, install/remove firewall rules). It does **not** own crypto, obfuscation, or status truth — helix-core is the only place WG + transport live (O2 [03-client §4.1]), and TunnelStatus is core-sourced (CI2). On Linux the core is linked **in-process** (desktop GUI / connector daemon) so the seam is a function-call boundary, not a socket — but the *ownership* split is identical to every other platform.



1. Non-negotiable Linux-shim invariants

Inherits the five TunnelPlatform obligations O1–O5 [03-client §4.1] and adds the Linux-specific ones below. A violation is a release blocker (§11.4 anti-bluff at the platform-shim layer; the shim is the only untyped seam — O5).

#	Invariant	Why / mechanical consequence
LX1	The shim never crypts/obfuscates ; it only creates the carrier device and hands the core a tun fd (userspace) or programs a kernel-WG peer (plain-udp).	O2 [03-client §4.1]; one core, no fork [01-TXP I4].

#	Invariant	Why / mechanical consequence
LX2	CAP_NET_ADMIN is acquired through exactly one sanctioned path (file-cap helper <i>or</i> polkit-gated helix-netd, §7) — never sudo from the GUI, never a setuid-root blob.	§11.4.161 rootless-first; §11.4.133 target-system safety.
LX3	startTunnel is idempotent + permission-aware: if privilege acquisition is denied, emit permissionDenied (not error), leave no half-open tun / orphan route / dangling nft table (§11.4.14).	O1 [03-client §4.1].
LX4	On stopTunnel and on every abnormal exit path the shim restores a quiescent host: tun destroyed, routes withdrawn, nft kill-switch table flushed <i>per the</i> §8.6/§11 <i>fail-closed rule</i> (kill-switch stays closed on crash, opens only on user-stop).	O4 [03-client §4.1]; O-I11 [01-ORC §0.3]; trap/Drop/systemd ExecStopPost.
LX5	The shim reports lifecycle as PlatformTunnelEvent (up/down/permissionDenied/revoked/error); it never synthesizes a TunnelStatus. Protection-state truth flows core→FFI→UI (CI2).	prevents "UI says connected while tunnel is down".
LX6	Backend choice is determined by the active Transport::kind() (§2), never guessed: plain-udp MAY use kernel WG; any obfuscated carrier MUST use the userspace tun path. The shim asks the core which backend the resolved rung needs; it does not infer.	§11.4.6 no-guessing; §2.3.
LX7	The connector daemon and the desktop GUI run the byte-identical core (--mode={client connector}), differing only in routing posture and privilege host (§9).	O-I13 [01-ORC §0.3], I4 [01-TXP].

2. The two data-path backends (kernel WG vs userspace tun)

The single most consequential Linux nano-detail. HelixVPN's obfuscation is **pluggable beneath WireGuard** [04_ARCH §2/§3, SYN §2]; helix-transport carriers (masque-h3, shadowsocks, udp-over-tcp, lwo, ...) sit *below* the WG datagram [01-TXP §0]. This collides with how Linux kernel WireGuard works.

2.1 Kernel WireGuard (netlink) — plain-udp ONLY

Kernel WG owns its **own** UDP socket to the peer endpoint; it encrypts in the kernel and sends WG datagrams directly out a raw UDP port [04_ARCH §3 table, SYN §5]. It is the fast path (zero userspace copies, no per-packet context-switch). It is configured over **netlink** (the wireguard genl family)

- rtnetlink for addresses/routes — via the wireguard-control / defguard_wireguard_rs + rtnetlink crates (UNVERIFIED exact crate choice — a Phase-1 dependency decision; the netlink *mechanism* is grounded [04_ARCH §5.3]).

Hard limitation (FACT, not a guess — §11.4.6): kernel WG speaks only plain WireGuard-over-UDP. It cannot route its datagrams *through* a userspace MASQUE / Shadowsocks / TCP obfuscator. Therefore **kernel WG is usable only when the active rung is plain-udp** (the unrestricted-network baseline [01-TXP §4.1]).

plain-udp via kernel WG:

[app traffic] → kernel route → wgX (kernel WG iface) → kernel encrypts
→ kernel UDP :51820 → peer (helix-core only PROGRAMS the peer; no per-packet pump)

In this mode helix-core's data loops (01-ORC §3) **do not pump packets** — the kernel does the datapath. The orchestrator still owns control: reconcile, kill-switch, DNS, status. It programs the kernel WG device (peer pubkey, endpoint, AllowedIPs) and reads last_handshake_time / rx/tx counters via netlink to drive Handshaking → Connected and PeerTimeout (§10, §13).

2.2 Userspace /dev/net/tun — every obfuscated carrier

When the active rung is **any** obfuscating transport, the datapath **MUST** be userspace: the shim opens /dev/net/tun, ioctl(TUNSETIFF, IFF_TUN|IFF_NO_PI) to create tunX, programs addresses/routes/MTU on it, and hands the **fd** to the core via attach_tun(fd) [03-client §3.1, 01-ORC §2.1]. The

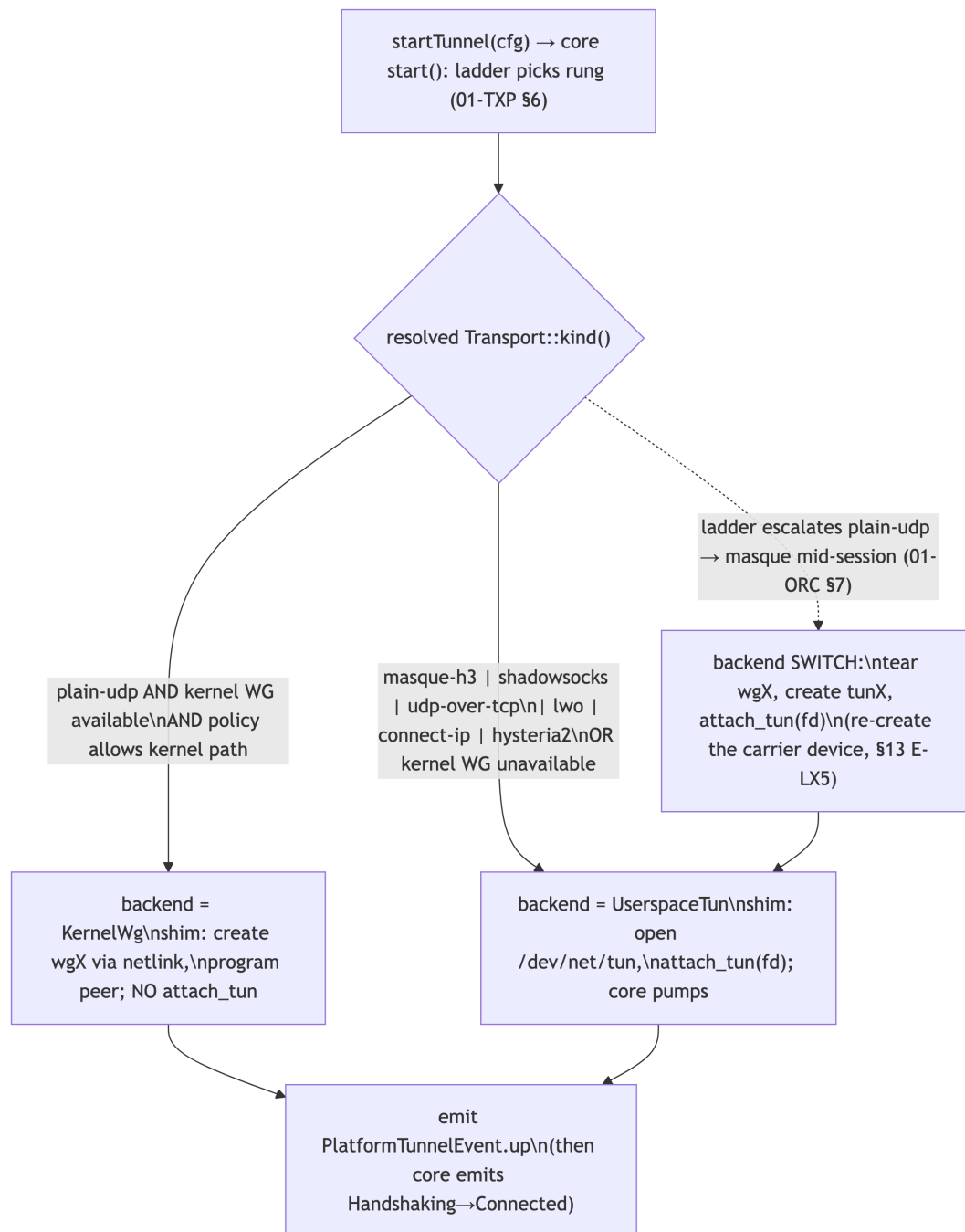
core's three loops [01-ORC §3] then run: plaintext IP $\leftarrow$ tun $\rightarrow$ helix-wg (boringtun userspace) encrypt $\rightarrow$ helix-transport obfuscate $\rightarrow$ OS socket; and the reverse.

masque-h3 (or any obfs) via userspace tun:

[app traffic] $\rightarrow$ kernel route $\rightarrow$ tunX (/dev/net/tun) $\rightarrow$ read by helix-core loop A
 $\rightarrow$ helix-wg encrypt $\rightarrow$ helix-transport (masque-h3) $\rightarrow$ QUIC/H3 :443 $\rightarrow$ gateway

This is the path that the iOS NE / Android VpnService shims use universally (they have no kernel-WG option) [03-client §5.1/§5.2]; on Linux it is the **obfuscated** path and the **fallback** path when kernel WG is unavailable (no wireguard module, missing CAP, container without the genl family) [04_ARCH §5.3 "boringtun userspace only as fallback"].

2.3 Backend selection (the core decides, the shim obeys — LX6)



Selection rule (closed, §11.4.6 — no guessing):

1. The core's ladder resolves the rung $\rightarrow$ `Transport::kind()` [01-TXP §6.3].
2. `kind() == "plain-udp"` **and** the host advertised `kernel_wg = true` at `startTunnel` **and** `KernelPathPolicy != ForceUserspace` $\Rightarrow$ **KernelWg**.
3. Otherwise $\Rightarrow$ **UserspaceTun** (the universal path; always available where `/dev/net/tun` + `CAP_NET_ADMIN` exist).
4. A **ladder escalation that changes kind()** across the kernel/userspace boundary (e.g. `plain-udp` blocked $\rightarrow$ escalate to `masque-h3`) is a **backend**

switch: the orchestrator's atomic carrier swap [01-ORC §7.3] is paired with a shim-side carrier-device swap — tear the kernel wgX, create the userspace tunX, `attach_tun(fd)`, re-program routes/DNS/kill-switch on the new device. The kill-switch stays **closed** across the swap (O-111) so no leak window.

Decision D-LXN-1 (surfaced, §11.4.66). Whether to ship the kernel-WG fast path in Phase 1 at all, or default every Linux session to the userspace tun for one uniform datapath (simpler, slightly slower for the plain-udp case). Option A (ship both): best plain-udp throughput on cooperative networks. Option B (userspace-only Phase 1): one datapath, no backend-switch complexity, matches iOS/Android exactly — defer kernel WG to Phase 2 as a perf optimisation.

Recommendation: B for Phase 1 (kill-switch/backend-switch correctness > peak plain-udp throughput; the userspace path is the one every obfuscated carrier needs anyway), promote A in Phase 2 behind a measured BENCH gain. See §15.

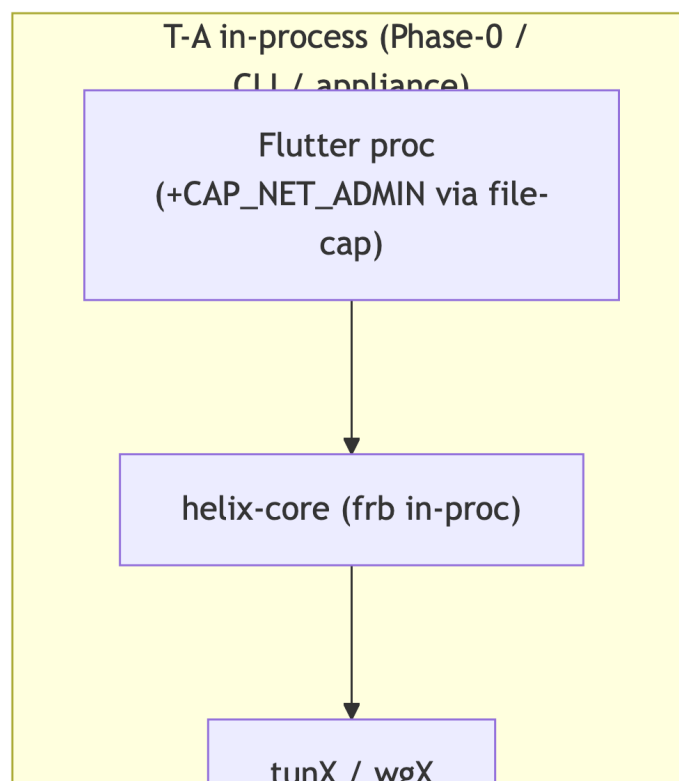
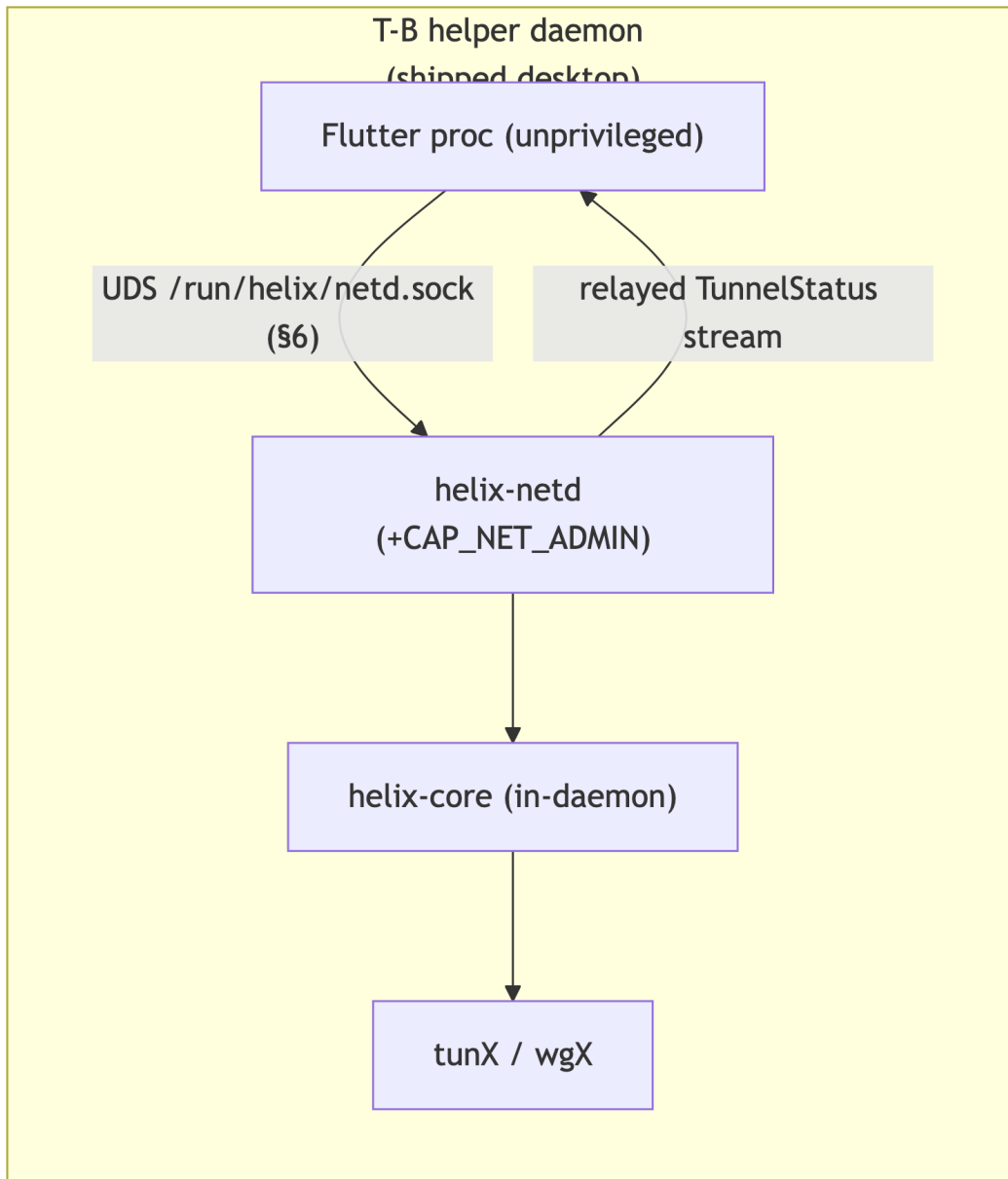
3. Deployment topologies (in-process vs privileged helper daemon)

Two ways the privileged datapath is reached from the unprivileged Flutter GUI [04_ARCH §5.3, 03-client §5.4]. Both implement the same Dart TunnelPlatform (§4); they differ only in *where CAP_NET_ADMIN lives*.

Topology	Core runs	Privilege held by	When	Mirror
T-A in-process	inside the Flutter process via frb	the Flutter process itself (file-caps on the launcher, §7.2) — or the app was launched by a service manager with caps	dev / single-user appliance / CLI; Phase-0 G5 reference [04_P0 §9]	(none — Linux-only simplification)
T-B helper daemon	inside a privileged helix-netd system service (§6); Flutter is unprivileged	helix-netd (systemd AmbientCapabilities=CAP_NET_ADMIN, polkit-gated)	desktop GUI on a multi-user workstation — recommended for shipped desktop	the Windows privileged service named-pipe model [03-client §5.3]

Why T-B for the shipped desktop GUI. Giving the whole Flutter+Skia process CAP_NET_ADMIN (T-A) widens the privileged attack surface to all of Dart + the UI toolkit. T-B confines privilege to a small audited Rust daemon (§6) and keeps the GUI unprivileged — the same reasoning that puts the core in a Windows service [03-client §5.3]. T-A remains the **Phase-0 spike** and the **CLI / headless appliance** topology where there is no separate GUI process to de-privilege [04_P0 §9].

In **T-A** the helix-core link is the frb in-process bridge [research-flutter_ffi; 03-client §3]; `attach_tun(fd)` is a direct call. In **T-B** the core lives in helix-netd; the Dart `LinuxTunnelPlatform` marshals `startTunnel/stopTunnel` over a Unix-domain socket (§6) and the **status stream crosses the socket** too — the GUI's frb binding subscribes to a relayed stream (§6.4), preserving CI2 (UI is a pure function of the core's status stream).



4. The TunnelPlatform contract — Linux impl

The Dart contract is fixed by [03-client §4] (one MethodChannel helixvpn/tunnel + one EventChannel helixvpn/tunnel/events, startTunnel/stopTunnel/events, TunnelConfig, PlatformTunnelEvent). On Linux the "channel" is **not** a Flutter platform channel to native code — there is no Kotlin/Swift layer — it is either an in-process frb call (T-A) or a UDS RPC (T-B). The Dart class is identical regardless:

```
// helix_core_ffi/lib/src/linux/linux_tunnel_platform.dart
// Implements the shared TunnelPlatform contract [03-client §4].
final class LinuxTunnelPlatform implements TunnelPlatform {
  LinuxTunnelPlatform(this._core, this._priv); // _priv =
    InProcPriv (T-A) | NetdClient (T-B)
  final HelixCore _core; // frb-
    generated core handle [03-client §3.2]
  final PrivilegedTunnelOps _priv; // §4.1 – the
    only privileged seam

  @override
  Future<void> startTunnel(TunnelConfig cfg) async {
    // 1. ask the privileged plane to acquire CAP_NET_ADMIN +
    //    decide backend (§2.3)
    final backend = await
      _priv.acquireAndSelectBackend(cfg); // throws
      PermissionDenied → §10
    // 2. create the carrier device under privilege; for userspace
    //    get the tun fd
    final created = await _priv.createCarrier(cfg,
      backend); // KernelWgHandle | TunFd
    // 3. start the core; userspace path hands it the fd (CI1:
    //    core never opens the tun)
    await _core.start(transport: 'auto', mapPathOrSession:
      cfg.sessionOrMapToken);
    if (created is TunFd) { await
      _core.attachTun(created.fd); } // §5
    // 4. lifecycle event (NOT status – LX5); status arrives via
    //    _core.statusStream()
    _emit(PlatformTunnelEvent(PlatformTunnelEventKind.up));
  }

  @override
  Future<void> stopTunnel() async {
    await _core.detachTun(); // userspace
    path; no-op on kernel path
    await _core.stop();
    await _priv.destroyCarrier(); // tear tun/
    wgX, withdraw routes (LX4)
    await _priv.revertKillSwitch(userInitiated: true); // §11 –
    the ONLY open path (§8.6)
    _emit(PlatformTunnelEvent(PlatformTunnelEventKind.down));
  }
}
```

```

}

@Override
Stream<PlatformTunnelEvent> events() => _events.stream; // up/
down/permissionDenied/revoked/error
}

```

4.1 PrivilegedTunnelOps — the privileged seam (the only Linux-specific surface)

```

// The abstraction that hides T-A (in-proc) vs T-B (helper daemon)
// from the rest of the app.
abstract interface class PrivilegedTunnelOps {
  /// Acquire CAP_NET_ADMIN (file-cap path: already held; helper
  /// path: polkit auth) and
  /// resolve the data-path backend for the ladder's chosen rung
  /// (§2.3). Throws
  /// PermissionDenied (→ permissionDenied event, LX3) if
  /// privilege is refused.
  Future<TunBackend> acquireAndSelectBackend(TunnelConfig cfg);

  /// Create the carrier device under privilege. KernelWg →
  /// program wgX peer (no fd);
  /// UserspaceTun → open /dev/net/tun, ioctl(TUNSETIFF), program
  /// addr/routes/MTU, return fd.
  Future<CarrierHandle> createCarrier(TunnelConfig
    cfg, TunBackend backend);

  /// Switch backend mid-session on a ladder escalation crossing
  /// the kernel/userspace
  /// boundary (§2.3 swap); kill-switch stays closed across the
  /// swap (0-I11).
  Future<CarrierHandle> switchCarrier(TunBackend to, TunnelConfig
    cfg);

  Future<void> destroyCarrier(); // idempotent
  (LX4)
  Future<void> applyKillSwitch(KillSwitchSpec s); // §11
  Future<void> revertKillSwitch({required bool
    userInitiated}); // §8.6 – open only if userInitiated
}

enum TunBackend { kernelWg, userspaceTun }
sealed class CarrierHandle {}
final class KernelWgHandle extends CarrierHandle { final String
  iface; KernelWgHandle(this.iface); }
final class TunFd extends CarrierHandle { final int fd;
  final String iface; TunFd(this.fd, this.iface); }

```

TunnelConfig is unchanged from [03-client §4]: overlayIp, routes, dnsServers, splitExcludeApps (per-route bypass on Linux — see §11), mtu (1280 default over MASQUE, 1420 plain WG [03-client §4]), sessionOrMapToken.

5. FFI surface consumed on Linux + the TUN handoff

The Linux shim consumes the **frozen** FFI surface [03-client §3.1] — it adds no new FFI verbs except the tun-fd handoff that already exists for Android (attach_tun/detach_tun) [03-client §3.1, 01-ORC §2.1]. Reproduced here for the Linux-relevant subset (consistency anchor — the canonical definition lives in [03-client §3.1]):

```
// helix-ffi/src/api.rs (subset consumed on Linux) – canonical
// surface: [03-client §3.1]
pub async fn start(cfg: ClientConfig) -> anyhow::Result<()>; //
// ClientConfig{ map_path_or_session, transport, mode }
pub async fn stop() -> anyhow::Result<()>;
pub fn status_stream(sink: StreamSink<TunnelStatus>); //
// §5.1
pub fn attach_tun(fd: i32) -> anyhow::Result<()>; //
// userspace path: hand the /dev/net/tun fd
pub fn detach_tun() -> anyhow::Result<()>; //
// userspace path teardown; no-op on kernel path
```

5.1 TunnelStatus — consistency with the data plane (LX5)

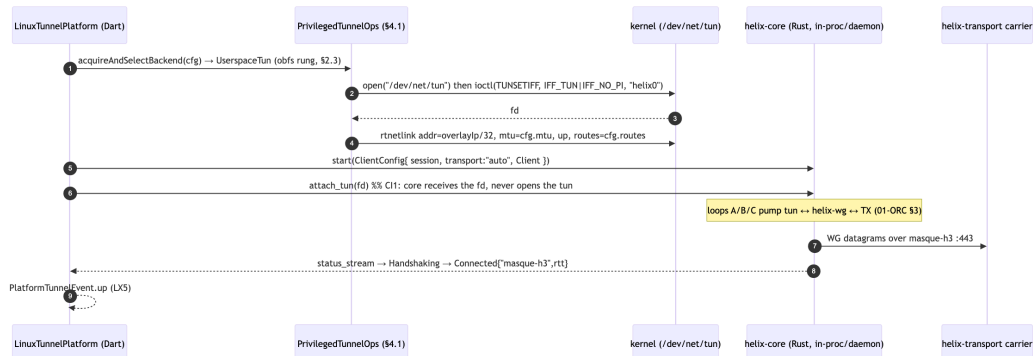
The **canonical data-plane status enum** is the 5-variant TunnelStatus emitted by the orchestrator over tokio::broadcast [01-ORC §4.1]:

```
// CANONICAL – [01-ORC §4.1]. The Linux shim NEVER constructs
// this; it is core-sourced.
pub enum TunnelStatus {
    Connecting,
    Handshaking,
    Connected { transport: String, rtt_ms:
        u32 }, // transport = Transport::kind() (e.g. "plain-
        udp", "masque-h3")
    Reconnecting,
    Down { reason: String }, // reason
    // prefix vocabulary: [01-ORC §4.4]
}
```

The **FFI mirror** [03-client §3.1] is a *UI superset* of the same enum — it adds Disconnected (the clean-idle projection of pre-start/stopped), a path field on Connected ("direct"|"relay", Phase 2 [03-client §3.1]), and Danger{kind} (leak/kill-switch-tripped, a UI projection driven by the §11 detector). Per [01-ORC §5.1], Idle/ShuttingDown map to Disconnected/Down at the FFI; Danger is *not* an orchestrator state — it is a UI overlay derived from a leak signal. **The Linux shim emits none of these**; it emits PlatformTunnelEvent only (LX5), and the orchestrator's broadcast reaches

Dart over frb (T-A) or the relayed UDS stream (T-B, §6.4). This keeps the FFI surface byte-consistent with [01-ORC §4.1] and [03-client §3.1]; a drift FAILs the FFI contract test [03-client §12].

5.2 The userspace tun-fd lifecycle (the one Linux-specific FFI flow)



fd ownership. After `attach_tun(fd)` the **core owns** the fd for read/write; the shim retains the fd only to `close()` it on teardown *after* `detach_tun()` returns (double-close guard: `detach_tun` makes the core drop its read loop and relinquish the fd; the shim then closes — never the reverse). `IFF_NO_PI` (no 4-byte packet-info prefix) is mandatory so the core's loop A reads raw IP packets [01-ORC §3.1] (UNVERIFIED: exact flags confirmed against the running kernel in the Phase-0 IT rig, not assumed).

6. The privileged helper daemon (helix-netd) + IPC

T-B's privileged plane. A small Rust daemon that **is** the privileged carrier manager + core host; the unprivileged Flutter GUI is a thin client. Mirrors the Windows `HelixTunnelSvc` + named-pipe design [03-client §5.3] with a Unix-domain socket instead of a named pipe.

6.1 Surface (the same TunnelPlatform verbs, over UDS)

```

// helix-netd/src/ipc.rs – length-prefixed CBOR (or protobuf)
// frames over a SOCK_SEQPACKET UDS.
// Socket: /run/helix/netd.sock (systemd socket-activated, §8).
// 0660 root:helix-netd group.
pub enum NetdRequest {
    StartTunnel { cfg: WireTunnelConfig }, // mirrors Dart
    TunnelConfig [03-client §4]
    StopTunnel,
    SetPin { kind: Option<String> }, // manual
    transport pin [01-ORC §2.1 set_pin]
}

```

```

ApplyMap { map: WireRouteMap },           // Phase 0 file-
      watch stand-in is daemon-side; Phase 1 stream
Subscribe,                                // opens the
      status relay (§6.4)
}
pub enum NetdResponse {
  Ok,
  PermissionDenied { detail: String },      // →
      permissionDenied event (LX3)
  Error { detail: String },                // honest reason
      (§11.4.6), never a secret (§11.4.10)

  Status(WireTunnelStatus),                // streamed
      after Subscribe (§6.4)
}

```

6.2 Authn / authz of the IPC

- The UDS is `0660 root:helix-netd`; only members of the `helix-netd` group (the GUI's runtime user, added at install) may connect — coarse gate.
- **Per-action authorization is polkit** (§7.3): `StartTunnel/StopTunnel` trigger the polkit action `digital.vasic.helixvpn.manage-tunnel`; the daemon calls polkit with the caller's `SO_PEERCREDS (pid/uid)` to authorize, so an interactive auth prompt is shown for a non-allowlisted user (`auth_admin_keep`) and silently allowed for an allowlisted desktop session (UNVERIFIED: exact polkit rule shipped — a Phase-1 packaging decision).
- The daemon **never** accepts a WG private key or a `SecretBytes` over the IPC in the clear beyond what the wire config already carries (§11.4.10); secrets in `WireTunnelConfig` are redacted in every daemon log (§13).

6.3 Single-owner discipline

`helix-netd` serves **one** active tunnel at a time (one carrier device, one core). A second `StartTunnel` while a tunnel is up is rejected `Error{busy}` (idempotency: a `StartTunnel` with the *same* cfg fingerprint is a no-op `Ok` — LX3). This is the §11.4.119 single-resource-owner rule applied to the host's tun device.

6.4 Status relay (preserving CI2 across the socket)



The relay applies the **same** receiver contract as the FFI [01-ORC §4.6]: Lagged → skip-to-latest (never an error), Closed → daemon gone → GUI shows Down{reason:"helper-unavailable"} and offers reconnect. Coalescing ([01-ORC §4.3]) happens core-side before the relay, so the socket carries only rare state-change frames.

7. Privilege model — CAP_NET_ADMIN, rootless, polkit, setcap

7.1 What needs privilege (FACT — §11.4.6)

Operation	Capability required	Used by
open("/dev/net/tun") + ioctl(TUNSETIFF) (create tun)	CAP_NET_ADMIN	userspace backend (§2.2)
Create/configure a kernel wireguard netlink device	CAP_NET_ADMIN	kernel backend (§2.1)
rtnetlink addr/route/MTU programming	CAP_NET_ADMIN	both backends
nftables kill-switch table install/ flush	CAP_NET_ADMIN	§11
Bind a low transport port (none — carriers bind ephemeral/:443-dest, not listen)	none (client dials outbound, SYN §1)	—

Honest boundary (§11.4.6). A **fully unprivileged** (zero-CAP) system-wide Linux tunnel is **not possible** — creating a TUN and programming routes both require CAP_NET_ADMIN; this is a kernel constraint, not a design choice (cf. §11.4.112 structural-impossibility class). UNVERIFIED: some distros allow a pre-created persistent tun owned by a group (tunctl/ip tuntap ... user) that an unprivileged process may then *open*, but route/firewall programming still needs CAP_NET_ADMIN — so the privilege cannot be fully eliminated, only *confined* (T-B). The §11.4.161 "rootless" mandate is satisfied by **never escalating to full root** — CAP_NET_ADMIN ≠ root, and the daemon drops every other capability (§8.2).

7.2 setcap / file-capability path (T-A, and the helper binary in T-B)

```
# install-time (packaging), the ONLY sanctioned grant (LX2):
setcap 'cap_net_admin,cap_net_raw=ep' /usr/lib/helixvpn/helix-
netd      # T-B helper
# (T-A appliance/CLI:) setcap 'cap_net_admin=ep' /usr/bin/
helixvpn-cli
```

cap_net_raw is added only if a carrier needs raw sockets (UNVERIFIED — none of the §2 carriers does today; included defensively for ICMP-based path probing, removed if unused per §11.4.124 dead-capability discipline). The GUI binary **never** carries file-caps in T-B (LX2).

7.3 polkit (T-B per-action gate)

A polkit action file `digital.vasic.helixvpn.manage-tunnel` (allow_active = auth_admin_keep by default; a desktop install MAY relax to yes for the console user via a shipped rule). `helix-netd` authorizes each `StartTunnel/StopTunnel` against the caller's `SO_PEERCRED`. Denial → PermissionDenied → permissionDenied event (LX3) — **never** an error (clean UX, [03-client O1]).

7.4 Rootless containerized path (connector, §11.4.161)

For the connector daemon shipped as a rootless Podman container (§9.4): the container is granted `--cap-add NET_ADMIN` and `--device /dev/net/tun` and runs as a non-root UID. UNVERIFIED: rootless Podman + `/dev/net/tun` + `NET_ADMIN` requires the host `tun` module loaded and the rootless user's `netns` to permit it — verified in the §11.4.76 containers-submodule on-demand IT rig, never assumed. The userspace `tun` backend (§2.2) is mandatory inside a container (kernel WG netlink to the *host* wg subsystem is generally not available to a rootless container `netns` — UNVERIFIED, measured in IT).

8. systemd integration

8.1 Desktop helper units (T-B)

```
# /usr/lib/systemd/system/helix-netd.socket — socket activation
    (daemon idle until first connect)
[Socket]
ListenStream=/run/helix/netd.sock
SocketMode=0660
SocketUser=root
SocketGroup=helix-netd
```

[Install]

WantedBy=sockets.target

/usr/lib/systemd/system/helix-netd.service

[Service]

```
Type=notify                                # sd_notify READY=1 once the
      IPC loop is serving (§8.3)
ExecStart=/usr/lib/helixvpn/helix-netd
AmbientCapabilities=CAP_NET_ADMIN          # the ONLY cap granted (LX2 /
      §7.1); NOT root
CapabilityBoundingSet=CAP_NET_ADMIN # nothing else is even
      reachable
NoNewPrivileges=yes
ProtectSystem=strict
ProtectHome=yes
PrivateTmp=yes
ProtectKernelModules=yes
ProtectControlGroups=yes
RestrictAddressFamilies=AF_UNIX AF_INET AF_INET6 AF_NETLINK    #
      UDS + sockets + netlink only
SystemCallFilter=@system-service
MemoryDenyWriteExecute=yes
ExecStopPost=/usr/lib/helixvpn/helix-netd --flush-killswitch-on-
      crash    # LX4 / 0-I11
```

ProtectSystem=strict etc. are §11.4.133 target-system-safety hardening; the daemon keeps **only** CAP_NET_ADMIN (§7.1). RestrictAddressFamilies includes AF_NETLINK (kernel-WG/route programming) and AF_INET/6 (the transport carriers dial outbound). ExecStopPost enforces fail-closed kill-switch on an abnormal stop (O-I11 [01-ORC §0.3]).

8.2 sd_notify lifecycle (the daemon systemd contract)

```
// helix-netd readiness + watchdog (Type=notify, WatchdogSec= set
      in §9 connector unit).
sd_notify(false, "READY=1\nSTATUS=serving on /run/helix/
      netd.sock");
// per loop: sd_notify(false, &format!
      ("WATCHDOG=1"));                                // liveness ping
// on reconfigure: sd_notify(false, "RELOADING=1"); ... "READY=1";
// on stop:        sd_notify(false, "STOPPING=1\nSTATUS=tearing
      down tunnel");
```

8.3 Why socket activation

The desktop helper is **idle (not running)** until the GUI first connects — zero resident cost when no tunnel is wanted. systemd starts helix-netd on the first UDS connection and (optionally) stops it after an idle timeout post-stopTunnel. For the **connector** (§9) socket activation is *off* — it is a persistent network appliance that must reconnect autonomously (fail-static, I3 [01-ORC §0.3]).

9. The connector-daemon deployment (--mode=connector)

The connector is the **same helix-core**, Mode::Connector — byte-identical loops, status stream, reconciler, transport ladder; only the routing posture differs (O-I13 [01-ORC §9], I4 [01-TXP]). Linux is its primary host [04_ARCH §5.4]. It is **headless** (no Flutter UI required; the Connector flavor GUI is optional config surface [03-client §6, D-CLIENT-4]).

9.1 What differs from the GUI client

Aspect	GUI client	Connector daemon
Topology	T-A or T-B (§3)	standalone privileged service (no GUI; core in the daemon process)
Carrier device	tun capturing the device default route	LAN iface + tun; forwards decapsulated packets into the served CIDR + NAT (no default-route capture) [01-ORC §9.1]
Kill-switch	Strict/ Permissive (privacy)	typically off — it is an appliance, not a privacy client (config knob, [01-ORC §9.4])
Inbound	n/a	outbound-only — dials the gateway, never listens (founding no-port-forward constraint [SYN §1])
Verdict map	n/a	installs the compiled policy as an nftables/eBPF verdict map , re-installed <1s on reconcile, default-deny fail-closed [01-ORC §9.2/§9.3]
Lifecycle host	GUI process / helper	systemd Type=notify service or rootless Podman quadlet (§9.4)

9.2 The connector data path (loop B branch)



The connector runs **loop A driven by the LAN iface** (not a TUN) and **loop B forwarding into the LAN** — TunSpec abstracts both so the loop bodies are unchanged [01-ORC §9.2]. CAP_NET_ADMIN is required for the tun, the LAN-forward NAT, and the nftables verdict map (§7.1).

9.3 systemd unit (connector)

```
# /usr/lib/systemd/system/helix-connectord.service
[Unit]
Description=Helix VPN Connector (outbound-only network agent)
After=network-online.target
Wants=network-online.target
[Service]
Type=notify
ExecStart=/usr/lib/helixvpn/helix-connectord --mode=connector --
    config /etc/helixvpn/connector.toml
AmbientCapabilities=CAP_NET_ADMIN
CapabilityBoundingSet=CAP_NET_ADMIN
NoNewPrivileges=yes
WatchdogSec=30 # §8.2 sd_notify WATCHDOG=1
    every <15s; restart on hang
Restart=on-failure
RestartSec=2 # bounded; complements the
    core's own backoff [01-ORC §7]
ProtectSystem=strict
ProtectHome=yes
PrivateTmp=yes
RestrictAddressFamilies=AF_UNIX AF_INET AF_INET6 AF_NETLINK
SystemCallFilter=@system-service
[Install]
WantedBy=multi-user.target
```

Restart=on-failure + WatchdogSec give the appliance autonomous recovery *on top of* the core's in-process reconnect ladder [01-ORC §7] — the systemd restart is the last-resort supervisor (E10 [01-ORC §12]: a panicked core is restarted; start() resets the ladder). The credential file /etc/helixvpn/connector.toml and any enrollment secret are 0600 root-owned and git-ignored (§11.4.10/.30) — never in the unit, never logged.

9.4 Rootless Podman quadlet (§11.4.76 / §11.4.161)

Per §11.4.76 all containerized deploy goes through the vasic-digital/containers submodule; the connector ships as an OCI image run by a Podman **quadlet** (.container unit managed by systemd, rootless) [04_ARCH §4, SYN §2]:

```
# ~/.config/containers/systemd/helix-connectord.container
    (rootless quadlet)
[Container]
Image=ghcr.io/vasic-digital/helix-connectord:helixvpn-<version>
    # §11.4.151 prefix
AddCapability=NET_ADMIN # ONLY this; rootless, non-
    root UID inside (§11.4.161)
AddDevice=/dev/net/tun
Exec=--mode=connector --config /etc/helixvpn/connector.toml
```

```

Notify=true
    passthrough
[Service]
Restart=on-failure
[Install]
WantedBy=default.target

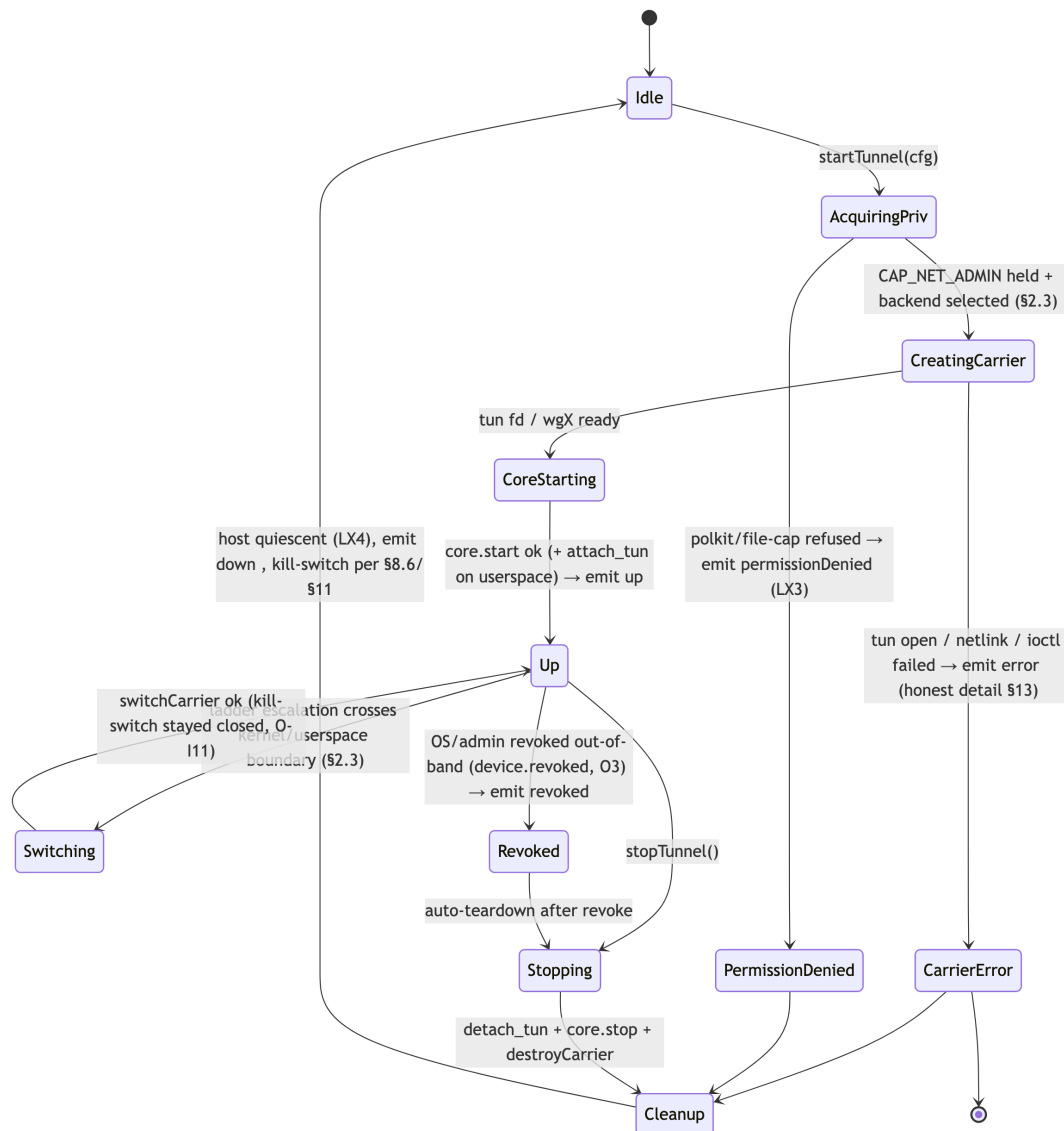
```

```
# quadlet → Type=notify
```

Inside the container the **userspace tun backend is mandatory** (§7.4); the container boots its integration infra on-demand via the containers submodule (§11.4.76 on-demand-infra invariant) for the IT test rig (§14).

10. Shim lifecycle state machine

The shim's **own** lifecycle (distinct from the core's OrchState [01-ORC §5]). It emits PlatformTunnelEvent (LX5), never TunnelStatus.



State→event map (the only events crossing to the UI per [03-client §4]):


```

    ip daddr <rfc1918> accept                                     #
allow_lan (config; printers etc.)
}
}

```

<gateway_endpoints> = KillSwitchConfig.allow_gateway_endpoints [01-ORC §8.2] — the carrier **MUST** always reach :443. The table is installed **closed before the first handshake** (Strict) so there is no pre-connect leak, and **stays installed on crash** (O-I11) — the systemd ExecStopPost (§8.1) only flushes on a *clean* user stop (§8.6). splitExcludeApps [03-client §4] maps on Linux to per-cgroup / per-fwmark accept rules (UNVERIFIED exact mechanism — Phase-2 split-tunnel item; the cgroup-net-cls / nftables meta cgroup path is the design reference).

11.2 DNS (the DnsApplyMethod matrix [01-ORC §8.3])

Host DNS stack	DnsApplyMethod	Mechanism
systemd-resolved present	systemd-resolved	resolvectl dns helix0 <tunnelDns> + resolvectl domain helix0 '~.' (route all)
resolvconf present	resolvconf	resolvconf -a helix0 with the tunnel servers
neither (static)	/etc/resolv.conf	atomic rewrite + restore-on-stop (backup the original)

block_plaintext_53 [01-ORC §8.3] adds nft rules dropping any off-tunnel :53 (UDP+TCP) when Connected — closing the classic DNS leak. The block **stays** on Reconnecting/Down (fail-closed) and reverts only on user-stop (§8.6, [01-ORC §8.4]). A DNS-apply failure at start is host-fatal → core emits Down{host-fatal} [01-ORC §8.2] — the shim never proceeds with an unverified DNS posture (§11.4.6).

12. Memory / size / performance budgets

Linux has **no iOS-NE-class hard ceiling** (CI4 [03-client §0.1] is iOS-only) — but the connector appliance / embedded host and the §11.4.161 rootless container warrant a real budget. Captured in the Phase-0 MEM/BENCH evidence, never asserted (§11.4.6/.123) [04_P0 §8].

Budget	Target	Basis
Kernel-WG backend userspace RSS	near-zero datapath (kernel owns packets);	§2.1 — fast path

Budget	Target	Basis
	core resident = control only	
Userspace-tun backend core RSS	low-tens MB steady-state (boringtun + transport buffers); no per-datagram heap growth	shares the helix-core budget [03-client §10]; reused scratch buffers [01-ORC §3.1]
helix-netd daemon RSS (idle, socket-activated)	0 (not running) until first connect (§8.3)	socket activation
helix-netd RSS (active, T-B)	core RSS + a thin IPC relay (single-digit MB overhead)	§6
Connector daemon RSS	low-tens MB; bounded by peers × Tunn + verdict map (no per-conn state, I5)	[01-ORC §13.2]
Through-tunnel throughput (plain-udp, kernel WG)	≥ 80% bare link (G1)	[04_P0 §8]; kernel datapath ≈ native WG
Through-tunnel throughput (userspace tun, masque-h3)	≥ 50% of plain-udp (G2); double-crypto/CC tax acknowledged	[01-TXP §11], [04_P0 §8]
Roam / iface-flap recovery	< 3 s	[01-ORC §7.5]; netlink iface watcher posts RoamDetected
Backend switch (kernel→userspace) latency	bounded; kill-switch closed throughout (no leak window)	§2.3, O-I11

The Linux iface watcher feeding RoamDetected [01-ORC §7.5] is **netlink** (RTM_NEWADDR/RTM_DELADDR on the rtnetlink route socket) — the shim's only ongoing background duty besides the carrier pump.

13. Error taxonomy & edge cases

Shim-layer errors map to PlatformTunnelEvent (§10); core errors flow through TunnelStatus ([01-ORC §10]). The shim's honest detail strings (§11.4.6) never carry a secret (§11.4.10).

Code	Cause	Event / mapping
LX-E1	polkit / file-cap refused	permissionDenied (NOT error, LX3)
LX-E2	/dev/net/tun absent (no tun module)	error{detail:"tun-module-missing"}; suggest modprobe tun (CLI)
LX-E3	ioctl(TUNSETIFF) / netlink addr/route failed	error + full host cleanup (LX4); no orphan iface
LX-E4	kernel wireguard genl family absent (kernel backend)	fall back to userspace tun (§2.3 rule 3), not an error
LX-E5	nft kill-switch install failed	core → Down{host-fatal} [01-ORC §8.2]; shim emits error
LX-E6	helix-netd UDS unreachable (T-B)	error{detail:"helper-unavailable"}; GUI offers retry/install

Edge cases (each → a §14 test point):

#	Edge case	Required behaviour
E-LX1	startTunnel called twice (same cfg)	idempotent no-op 0k (LX3); different cfg while up → Error{busy} (§6.3)
E-LX2	privilege granted but tun create fails	full cleanup, no half-open device (LX4); error
E-LX3	core crashes while shim holds the tun fd	T-A: process dies, kill-switch rules persist (O-I11), systemd/supervisor restarts; T-B: daemon ExecStopPost flush-on-crash leaves closed rules
E-LX4	detach_tun then shim closes fd, but a core read loop lingered	ordering guard (§5.2): core drops the loop on detach_tun <i>before</i> shim close(); double-close impossible
E-LX5	ladder escalates plain-udp(kernel)→masquerade3(userspace) mid-session	backend switch (§2.3): tear wgX, create tunX, attach_tun, re-program routes/DNS/KS; kill-switch stays closed across the swap (O-I11); recovery within the [01-ORC §7] budget
E-LX6	host has no systemd (e.g. minimal container/Alpine-OpenRC)	T-A path only; helper-daemon socket-activation degrades to a plain spawned daemon; UNVERIFIED non-systemd init support is a Phase-2 packaging item

#	Edge case	Required behaviour
E-LX7	device.revoked arrives out-of-band (admin kills the connector)	core → Down{auth-failed}; shim emits revoked (O3); kill-switch stays closed (connector Off → traffic just stops forwarding)
E-LX8	rootless container lacks NET_ADMIN	startTunnel → permissionDenied with detail:"container-missing-net-admin"; documented remediation (§7.4) — never a fake success

14. Test points (§11.4.169)

Every Linux-shim workable item declares its required test types from the §11.4.169 closed vocabulary [06-phase0-spike-wbs §0.4]; the ONLY permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason, never a silent gap. Four-layer enforcement per §11.4.4(b) on every closure. Every PASS ships rock-solid captured evidence (§11.4.5/.69/.107) — config-only / grep-only PASS forbidden.

Test id	§11.4.169 types	What it proves	Rig / captured evidence
LX-1	UT	PrivilegedTunnelOps backend-selection table (§2.3) is total + correct for every kind() × kernel_wg× policy	table-driven Rust/Dart unit test; mocks allowed (unit only, §11.4.27)
LX-2	UT	nft kill-switch ruleset render is exact for Strict/Permissive/Off + allow_lan/allow_gateway	golden nft-script diff
LX-3	IT	userspace path: open /dev/net/tun, attach_tun(fd), loop A/B move real WG datagrams; curl http://10.10.0.20/ over plain-udp and masque-h3	netns + edge + connector booted via containers submodule (§11.4.76); tcpdump + curl 200
LX-4	IT	kernel-WG path: program wgX via netlink, last_handshake_time drives Connected; curl over plain-udp	netns + kernel wireguard; netlink dump + curl 200
LX-5	E2E+FA	full status arc Connecting→Handshaking→Connected{plain-udp,rtt} driven ONLY by the core stream on Flutter-Linux (G5)	window-scoped MP4 + vision/OCR verdict (§11.4.159) of StatusChip

Test id	§11.4.169 types	What it proves	Rig / captured evidence
LX-6	E2E+FA	backend switch plain-udp(kernel)→masque-h3(userspace) on a DPI block; zero leak across the swap (E-LX5)	reading plain-udp · 23ms nft block plain WG mid-session; tcpdump underlay shows ZERO app packets during swap; ordered status trace (N=3 identical, §11.4.50)
LX-7	SEC	kill-switch closes on drop — no plaintext egress while Reconnecting/Down; closed rules persist on simulated crash (E-LX3)	block gateway mid-Connected; tcpdump underlay = ZERO leaked packets; SIGKILL the core, assert nft table still present
LX-8	SEC	DNS-leak guard: off-tunnel :53 dropped when Connected; system DNS restored on user-stop only (§11.2)	dig @off-tunnel blocked; resolvectl/ resolv.conf capture before/after
LX-9	SEC	privilege confinement: helix-netd holds only CAP_NET_ADMIN; GUI binary carries no file-caps (LX2); secrets absent from daemon logs (§11.4.10)	getpcaps/capsh dump; getcap on GUI; plant-a-secret grep-empty proof
LX-10	E2E	polkit denial → permissionDenied (NOT error), no half-open tun (LX3/E-LX2)	deny the polkit action; assert event kind + ip tuntap show empty
LX-11	INT	connector posture: gateway→connector→LAN forward + reverse, ACL-gated default-deny; verdict map re-install <1s on reconcile	netns LAN; authorized host reachable, unauthorized DROPPED+counted; pid unchanged on policy edit
LX-12	SC	iface flap → RoamDetected (netlink) → re-dial → Connected <3s; process-kill mid-transfer → systemd restart recovers	netns flap + SIGKILL; status-trace timing CSV; recovery <3s

Test id	§11.4.169 types	What it proves	Rig / captured evidence
LX-13	CONC+RACE	concurrent startTunnel/stopTunnel + backend switch under load drop zero datagrams; UDS relay no deadlock	concurrency harness; cargo test --features loom on the swap; relay log
LX-14	MEM+BENCH	kernel-WG ≥80% bare link (G1); userspace masque ≥50% of plain (G2); core RSS bounded, no per-datagram heap growth	iperf3 via bench.sh CSV; /proc/<pid>/status RSS sample
LX-15	LOAD	connector under 1/10/100 concurrent client sessions + handshake-flood: no fd leak, bounded mem, verdict map holds	iperf3 fan-in + handshake-flood CSV; ls /proc/<pid>/fd count stable
LX-16	CH+HQA	a challenges/helix_qa bank entry scores PASS only on the captured LX-5/LX-6/LX-7 evidence (not config)	HelixQA autonomous session, evidence-gated (§11.4.27/.107); Challenge result.json

Each gate ships a paired §1.1 meta-test mutation so it provably cannot bluff — e.g. flip the shim to `close()` the tun fd *before* `detach_tun` returns → LX-13 double-close guard FAILs; weaken the connector verdict gate → LX-11 FAILs; emit up before privilege is acquired → LX-10 FAILs; flush the kill-switch on crash → LX-7 FAILs.

15. Open decisions surfaced by this document

Per §11.4.6/§11.4.66 — options + recommendation, never silently resolved.

#	Decision	Option A	Option B	Recommendation
D-LXN-1	Kernel-WG fast path in Phase 1?	ship both backends (best plain-udp throughput)	userspace-tun-only Phase 1 (one datapath, no swap complexity, matches	B for Phase 1; promote A in Phase 2 behind a measured BENCH gain (§2.3).

#	Decision	Option A	Option B	Recommendation
			iOS/ Android)	
D- LXN-2	Desktop GUI privilege topology	T-A in-process (file- caps on the GUI)	T-B helper daemon helix-netd (polkit, confined caps)	T-B for the shipped GUI (small audited privileged surface, mirrors Windows); T-A for Phase-0 spike + CLI/ appliance (§3).
D- LXN-3	netlink crate for kernel WG + routes	defguard_wireguard_rs + rtnetlink	hand- rolled genl/ rtnetlink	A (reuse a maintained crate) per §11.4.74 catalogue-first; UNVERIFIED maturity — confirm in Phase-1 spike before committing.
D- LXN-4	Split-tunnel (splitExcludeApps) mechanism	cgroup-net-cls + fwmark + nft meta cgroup	network- namespace per app	A (Phase 2) — lower blast radius, no per-app netns; surfaced, not built in MVP (§11.1).
D- LXN-5	Non-systemd init support (OpenRC/ Alpine)	T-A spawned daemon only	full OpenRC service scripts	A for MVP , B as a Phase-2 packaging item (E-LX6); UNVERIFIED demand.

16. Cross-document contracts this document consumes/fixes

Contract	Source of truth	This doc's role
FFI surface (start/stop/ status_stream/ attach_tun/detach_tun)	[03-client §3.1]	consumes unchanged; adds no FFI verb (§5)
TunnelStatus (5-variant canonical)	[01-ORC §4.1]	consumes ; shim never constructs it (LX5); FFI

Contract	Source of truth	This doc's role
		mirror superset noted (§5.1)
TunnelPlatform channel contract (startTunnel/stopTunnel/events, TunnelConfig, PlatformTunnelEvent)	[03-client §4]	implements for Linux (§4); fixes the PrivilegedTunnelOps Linux seam
Transport::kind() → backend selection	[01-TXP §2/§6]	consumes to choose kernel-WG vs userspace tun (§2.3, LX6)
Kill-switch/DNS side-effect matrix	[01-ORC §8]	applies on Linux via nftables/resolved (§11)
Connector posture (Mode::Connector)	[01-ORC §9]	fixes the Linux daemon deployment (§9): systemd + rootless Podman quadlet
CAP_NET_ADMIN / rootless / containers	§11.4.76/.161/.133, [SYN §2]	fixes the privilege model (§7) + quadlet deploy (§9.4)

Sources verified

- final/03-client-core-and-ui.md [03-client] — §0.1 (CI1/CI2/CI4 invariants), §3 (FFI surface + TunnelStatus FFI mirror), §4 (TunnelPlatform contract, obligations O1–O5), §5.3 (Windows service/pipe parallel), §5.4 (Linux row), §6 (flavors), §10 (budgets), §12 (testing).
- final/v02-data-plane/orchestrator-and-state.md [01-ORC] — §0.3 (O-I10/11/13), §2.1 (attach_tun, Mode, TunSpec), §3 (three loops), §4.1 (canonical TunnelStatus), §4.4 (Down.reason), §5 (state machine), §7 (backoff/roam), §8 (kill-switch + DNS apply methods), §9 (connector posture/verdict map), §12 (edge cases), §13.2 (perf budgets).
- final/v02-data-plane/transport-trait.md [01-TXP] — §0/§2 (Transport, kind()), §4 (per-carrier dial: plain-udp vs obfuscated), §6 (ladder/dial), §11 (throughput budgets G1/G2).
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md [04_ARCH] — §3 (WG crypto core + pluggable transports), §5.3 (per-platform shim matrix; Linux = kernel WG/tun, Rust), §5.4 (connector daemon), kernel-vs-userspace WG note, Podman quadlets.
- 04_VPN_CLD/HelixVPN-Phase0-Spike.md [04_P0] — §9 (G5 Flutter-Linux FFI demo; "on Linux the core can drive the TUN directly"), §8 (gates / recovery <3s).

- 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md [04_UI] — §6 (shim contract: shim does only configure/handoff/lifecycle).
- v09-research/_SYNTHESIS.md [SYN] — §1 (outbound-only), §2 (rootless Podman, kernel-WG-fast-path/boringtun-fallback), §5 (client architecture), §9 (constitution bindings).
- 06-phase0-spike-wbs.md §0.4 — the §11.4.169 test-type code map (UT/IT/E2E/FA/ CH/HQA/LOAD/SEC/SC/CONC/RACE/MEM/BENCH).
- [research-flutter_ffi] — generic flutter_rust_bridge v2 in-process binding pattern (as grounded by [03-client §3]); [research-ios_android] — native-shim precedent (Linux in-process contrast). Items not in the read evidence base are tagged UNVERIFIED per §11.4.6 (kernel-WG crate choice, IFF_NO_PI flags, rootless-container tun reachability, polkit rule, non-systemd init, split-tunnel cgroup mechanism).

Constitution: §11.4.6 (no-guessing / UNVERIFIED tags), §11.4.10 (secret redaction), §11.4.66 (decisions = options + recommendation), §11.4.76 (containers submodule), §11.4.112 (structural-impossibility honesty — zero-CAP tunnel), §11.4.119 (single-resource-owner), §11.4.124 (dead-capability discipline), §11.4.133 (target-system safety / systemd hardening), §11.4.151/.155 (release/ recording prefixes), §11.4.159 (window-scoped vision-verified evidence), §11.4.161 (rootless), §11.4.169 (test-type coverage), §1.1 (paired mutations).

End of nano-detail specification — Linux shim (kernel WG / tun), Volume 4 (Clients). Pairs with [03-client §5.4] (the row this deepens), [01-ORC] (the core it hosts in-process), [01-TXP] (the carrier whose kind() drives backend selection), and the sibling shim docs (apple / android / windows / harmonyos / aurora). Surfaced decisions: D-LXN-1..5 (§15) — presented, none silently resolved (§11.4.66).